

Two computational diagnostics for Hecke L -functions over $\mathbb{Q}(\omega)$, and a sampling artifact in angular statistics

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Abstract

We report two practical diagnostics arising from a systematic numerical study of Hecke L -functions attached to the Eisenstein field $K = \mathbb{Q}(\omega)$, $\omega = e^{2\pi i/3}$: cubic residue characters, S_3 Artin L -functions of pure cubic fields, and Grössencharaktere of infinity type $k - 1$ for $2 \leq k \leq 7$. Neither the underlying arithmetic nor the identities verified here are new; all are classical (Hecke; Eichler; Andrianov; Ireland–Rosen). The contribution is methodological. First, we observe that solving for the root number ε from the approximate functional equation *without* imposing $|\varepsilon| = 1$, and then testing whether $|\varepsilon| = 1$ holds, yields a stringent simultaneous check on the conductor, the gamma factor, and every ramified Euler factor; in our computations it detected four independent errors, three of them a missing Euler factor at the ramified prime $\lambda = 1 - \omega$. Second, we document a sampling artifact: canonical Cornacchia output confines $\arg \pi$ to a quadrant, and the subsequent unit normalisation injects a spurious harmonic of order exactly 6 into angular statistics — measured here at 69.7σ — which is invisible in the Frobenius traces a_p but corrupts any statistic sensitive to $\arg \pi$. We give the corrected enumeration and confirm Hecke equidistribution and the arcsine law for the normalised traces.

1 Setting and disclaimer of novelty

Let $K = \mathbb{Q}(\omega)$ with $\omega^2 + \omega + 1 = 0$, $\mathcal{O}_K = \mathbb{Z}[\omega]$, class number one, unit group $\mathcal{O}_K^\times \cong \mathbb{Z}/6$ generated by $\zeta_6 = -\omega^2$, and $\lambda = 1 - \omega$ the prime above 3, with $(3) = (\lambda)^2$. Write $N(a + b\omega) = a^2 - ab + b^2$. An element is *primary* if $\alpha \equiv 1 \pmod{3}$; each ideal coprime to 3 has a unique primary generator.

For an integer $k \geq 2$ define the Grössencharakter

$$\psi_k(\mathfrak{a}) = \alpha^{k-1}, \quad \alpha \text{ the primary generator of } \mathfrak{a}, \quad (1)$$

so that $L(s, \psi_k)$ is the L -function of a CM newform of weight k . All of the following are classical and are used here only as ground truth:

- CM forms as theta series with spherical harmonics, and the action of Hecke operators on such theta series (the *Eichler Commutation Relation*); see Andrianov [1] and the references there. In rank 2 with class number one this reduces to the statement that $\{N = pn\} = \pi\{N = n\} \cup \bar{\pi}\{N = n\}$ with overlap $p\{N = n/p\}$, whence $a_{pn} = a_p a_n - p^{k-1} a_{n/p}$.
- Hecke operators as lattice correspondences (Shimura [5], Ch. 3).
- Equidistribution of $\arg \pi$ for primes of an imaginary quadratic field (Hecke [2]).
- The ramification criterion for $K(\sqrt[3]{\alpha})/K$ and the cubic supplementary law (Ireland–Rosen [3], Ch. 9).

We claim no new theorems. The two items in Sections 3 and 4 are practical observations that we have not found stated explicitly in the literature and that materially affected the reliability of our computations.

2 The unit group controls the conductor

Since ψ_k in (1) must be well defined on ideals, it is required that $u^{k-1} = 1$ for every unit u in the relevant congruence class. The stabiliser subgroup of $\mathbb{Z}/6 \cong \mathbb{Z}/3 \times \mathbb{Z}/2$ therefore fixes the conductor $\mathfrak{f}$, and the level of the associated newform is $|d_K| N(\mathfrak{f}) = 3N(\mathfrak{f})$:

k	$k-1$	stabiliser	$\mathfrak{f}$	level
2	1	trivial	(3)	27
3	2	$\mathbb{Z}/2$	(3)	27
4	3	$\mathbb{Z}/3$	(λ)	9
5	4	$\mathbb{Z}/2$	(3)	27
6	5	trivial	(3)	27
7	6	$\mathbb{Z}/6$	(1)	3

Remark 1. *The $\mathbb{Z}/2$ factor $\{\pm 1\}$ occurs as a stabiliser but never yields its own conductor: $-1 \equiv 1 \pmod{\mathfrak{m}}$ forces $\mathfrak{m} \mid 2$, and 2 is inert in $\mathbb{Z}[\omega]$, so no ideal in the tower $(1) \supset (\lambda) \supset (3)$ realises it.*

Remark 2 (a guard). *The multiplicative group $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$ has order 2, not 6: the 2-part is trivial. There are exactly two Dirichlet characters modulo 6, of conductors 1 and 3. In particular no character modulo 6 has even conductor, so modulus 6 carries no discriminant -4 content. The $\mathbb{Z}/3 \times \mathbb{Z}/2$ decomposition above is a statement about units, not about residue classes of primes; the additive isomorphism $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is true but does not transfer.*

The levels were verified against three independent anchors: $k = 2$ against the elliptic curve $y^2 + y = x^3$ of conductor 27 (all $p < 400$); $k = 4$ against $\eta(3\tau)^8$ (all $p < 4000$, exact integer arithmetic); and $k = 3$ against the theta series $\sum_{\alpha \text{ primary}} \alpha^2 q^{N(\alpha)}$ computed by lattice enumeration (max discrepancy 1.9×10^{-12} for $n \leq 3000$).

3 Diagnostic I: the unconstrained root number

Let $\Lambda(s) = (\sqrt{N}/2\pi)^s \Gamma(s) L(s)$ with functional equation $\Lambda(s) = \varepsilon \Lambda(k-s, \bar{\chi})$ and smoothed approximate functional equation

$$\Lambda(s) = \sum_{n \geq 1} a_n \left(\frac{\sqrt{N}}{2\pi n} \right)^s \Gamma\left(s, \frac{2\pi n}{\sqrt{N}}\right) + \varepsilon \sum_{n \geq 1} \bar{a}_n \left(\frac{\sqrt{N}}{2\pi n} \right)^{k-s} \Gamma\left(k-s, \frac{2\pi n}{\sqrt{N}}\right). \quad (2)$$

The procedure. Compute $L(s_0)$ at a real point s_0 in the region of absolute convergence by direct summation of the Dirichlet series. Solve (2) for ε as a single complex unknown, *imposing no constraint*. Then test $|\varepsilon| = 1$, and independently verify (2) at a second point $s_1 \neq s_0$.

Why it is stringent. Both $|\varepsilon| = 1$ and the s_1 identity are consequences of the functional equation that the solve does not enforce. An error in the conductor N , in the gamma factor, or in any single Euler factor — including at ramified primes — destroys them. In practice we observed $||\varepsilon| - 1| \lesssim 10^{-10}$ when everything is correct and $\gtrsim 10^{-2}$ otherwise, a separation of eight orders of magnitude.

Remark 3 (a caution). *It is essential that the two summands in (2) use a_n and $\bar{a}_n$ respectively. If both use a_n , the identity $\Lambda(s) = \varepsilon \Lambda(k-s)$ becomes a tautology satisfied by arbitrary coefficients: we verified that random integer sequences pass with residual exactly 0. A self-consistency check of this kind is worthless; the test must compare against an independently computed value.*

Sample separations. For the S_3 Artin L -function of $\mathbb{Q}(\sqrt[3]{2})$ (degree 2, weight 1):

$$\begin{array}{llll} N : & 21 & 63 & \mathbf{189} & 567 \\ |\varepsilon| : & 1.5048 & 1.2428 & \mathbf{0.9999999959} & 0.3567 \end{array}$$

For the cubic residue character $(\pi_7/\cdot)_3$ of conductor 189, ε solved unconstrained returned $\varepsilon = 0.89595322 - 0.44414843i$ with $|\varepsilon| = 1.0000000000$ and the second identity verified to 3.9×10^{-13} ; a complex root number on the unit circle emerging from an unconstrained linear solve is a strong simultaneous certificate.

Errors detected. In the course of this work the test flagged four independent errors, each of which had passed inspection:

Error	Symptom	Correction
Missing Euler factor at λ , $k = 7$	no valid N among 3, 9, 27	$a_3 = \lambda^6 = -27$, level 3
Same, $\mathbb{Q}(\sqrt[3]{m})$, $m \equiv \pm 1 \pmod{9}$	$ \varepsilon = 0.69\text{--}0.85$	$a_3 = +1$, cond. exponent 1
Incomplete conductor rule	level 9 predicted where 3 holds	three regimes, not two
Twist conductor	predicted 189, $ \varepsilon = 0.66$	$\mathfrak{f} = (\lambda\pi_7)$, level 63

The third row is worth isolating. The condition “ $3 \mid k - 1$ ” is necessary but not sufficient for level 9: when $6 \mid k - 1$ the stabiliser is all of $\mathbb{Z}/6$, the conductor is trivial, and the level drops to 3. Equivalently, λ^{k-1} is a rational integer precisely when $6 \mid k - 1$ ($\lambda^6 = -27$, $\lambda^{12} = 729$), which is exactly when λ becomes unramified and acquires an Euler factor. The two statements are the same statement.

Determining unknown local data. Where λ is unramified, the cubic residue symbol $(\alpha/\lambda)_3$ cannot be computed from the power formula, since the residue field $\mathbb{F}_3$ contains no primitive cube root of unity. The $|\varepsilon| = 1$ test determines it uniquely: for $N(\alpha) = 19, 37, 73$ the three candidate values $\omega^0, \omega^1, \omega^2$ gave $|\varepsilon| \in \{1.4478, 1.1162, \mathbf{1.0000000000}\}$ and permutations, with exactly one admissible in each case. The resulting values satisfy $(\alpha/\lambda)_3 = \omega^{c \bmod \lambda}$ where $\alpha = 1 + 3\lambda c$; this agrees with the directly computable $(\lambda/\alpha)_3$ on 2996/2996 hyperprimary primes of norm $< 200,000$, and is the classical supplementary law [3].

4 Diagnostic II: a sampling artifact in angular statistics

Proposition 1 (empirical). *Let π_p be obtained by applying Cornacchia’s algorithm to $p \equiv 1 \pmod{3}$ and then rotating by the unique unit making the result primary. Then the angles $\theta_p = \arg \pi_p$ are not equidistributed: they carry a harmonic of order exactly 6.*

Cornacchia returns $p = a^2 + 3b^2$ with $a, b \geq 0$; the associated element $z = a + i\sqrt{3}b$ has $\arg z \in [0, \pi/2]$ always. The primary normalisation then rotates by a multiple of $\pi/3$. A window of width $\pi/2$ tiled at spacing $\pi/3$ covers the circle non-uniformly with period exactly $\pi/3 = 2\pi/6$. Compounding this, a single call returns only one of the two conjugate prime ideals above p .

Measurement. With 108,339 split primes below 3×10^6 :

Statistic	via Cornacchia+primary	direct lattice enumeration
$N^{-1} \left \sum e^{i\theta} \right $	0.001858	0.000464
$N^{-1} \left \sum e^{6i\theta} \right $	0.211786 (69.7 σ)	0.000905 (0.42 σ)
$N^{-1} \left \sum e^{7i\theta} \right $	0.000948	0.000040
Kolmogorov–Smirnov $\sqrt{n} D$	9.564	0.300

(Critical value at 5% is 1.36.) The corrected enumeration lists all primary α of prime norm directly, yielding $216,678 = 2 \times 108,339$ elements, i.e. both conjugates.

Remark 4. *The artifact is invisible in $a_p = \pi + \bar{\pi}$, which is symmetric in the two conjugates and independent of the unit choice. Every trace-based computation is therefore unaffected. It corrupts only statistics sensitive to $\arg \pi$ — which is precisely the angular equidistribution one wishes to test.*

5 Confirmations after correction

With the corrected enumeration, Hecke equidistribution holds (Section 4), and the normalised traces $a_p/(2p^{(k-1)/2}) = \cos((k-1)\theta)$ follow the arcsine law $\frac{1}{\pi\sqrt{1-x^2}}$, as they must when θ is uniform:

	$E[x^2]$	$E[x^4]$	$E[x^6]$
CM, weight 2 (measured)	0.49946	0.37437	0.31183
arcsine (predicted)	0.50000	0.37500	0.31250
37a, non-CM (measured)	0.24915	0.12336	0.07636
semicircle (predicted)	0.25000	0.12500	0.07812

Weights $k = 2, 3, 4, 7$ all give L^1 distance ≈ 0.031 to the arcsine density and ≈ 0.303 to the semicircle. The measures differ most sharply at the edge: the CM density diverges at $x = \pm 1$ (measured 1.4332 on $|x| > 0.9$) while the non-CM density vanishes there (measured 0.1767).

6 Computational remark

Because the primary class $a \equiv 1$, $b \equiv 0 \pmod{3}$ is a sublattice of index 9, computing $\{a_n\}_{n \leq X}$ by direct lattice enumeration touches $(2\pi/\sqrt{3})X/9 = 0.403 X$ points (measured at 0.403 for X up to 2×10^6), requires no primality sieve and no modular square roots, and ran $4\text{--}8\times$ faster than the sieve-plus-Cornacchia route in our implementation. The comparison reverses completely for a single large prime, where Cornacchia is polylogarithmic (0.116 ms at $p \approx 2^{60}$) and enumeration is hopeless.

Remark 5. *Summing α^{k-1} over all lattice points, without the congruence restriction, gives identically zero for $k = 3$, since $\sum_{u \in \mathcal{O}_K^\times} u^2 = 2(1 + \omega + \omega^2) = 0$. The congruence class is not a normalisation convenience.*

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